Continuity Burden in Longitudinal Human-AI Interaction: An Empirical Case Study of Emotionally-Indexed Persistent Memory

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Abstract

This paper introduces and quantifies continuity burden—the communicative effort humans expend re-establishing context when interacting with memoryless AI systems. Drawing on 66,380 messages across 825 conversations conducted without persistent memory, we operationally define continuity references and find a rate of 4.8 per 1,000 messages, with frequency significantly associated with conversation length ($R^2 = 0.381$, $p < 0.001$): 59.3% of conversations exceeding 200 messages contain continuity references, compared to 2.1% of conversations under 10 messages. References are dominated by frustration markers (56.6%) and decline over time, consistent with user adaptation. We further document the operational characteristics of the Circular Associative Memory Architecture (CAMA), an emotionally-indexed persistent memory system comprising 52,602 memories with 99.98% affect coverage, and discuss its potential as an architectural mechanism for addressing continuity burden. As a single-participant longitudinal case study, these findings are exploratory and require multi-participant replication.

Keywords: human-AI interaction, emotionally-indexed memory, continuity burden, longitudinal case study, affective computing, memory architecture

1. Introduction

Large language models operate without persistent memory across conversation boundaries. Each session begins from a blank state, requiring the human participant to re-establish context, re-explain preferences, and re-build rapport. In extended human-AI relationships spanning months or years, this architectural limitation may impose a communicative cost on the human participant—a cost we term continuity burden. This paper provides the first empirical quantification of continuity burden in long-term human–AI interaction.

The Circular Associative Memory Architecture (CAMA) was designed to address this limitation through three structural layers. The first layer (Shelves) provides persistent storage with emotional annotations—each memory carries valence, arousal, and discrete emotion chord values alongside its content. The second layer (Racks) maintains relational connections between memories indexed by emotional and semantic similarity rather than chronology. The third layer (Console) implements a circular working memory ring of fixed capacity, where a subset of active memories persists across conversation boundaries to support partial state continuity. The architecture, its theoretical foundations, and its engineering implementation have been described in three prior publications (Reinhold, 2026a; 2026b; 2026c). This paper provides an empirical evaluation of CAMA’s operational characteristics using longitudinal interaction data from a single researcher’s extended engagement across multiple AI platforms.

The pre-CAMA and post-CAMA conditions do not represent separate temporal periods. Dataset B (66,380 raw messages) and Dataset A (52,602 CAMA memories) draw from overlapping interaction periods, both beginning January 2025. The condition distinction is architectural, not temporal: Dataset B represents interaction without persistent memory infrastructure, while Dataset A represents the same and additional interaction history processed through CAMA’s emotional indexing and retrieval system. CAMA became operationally active in March 2026. The participant is also the system designer, which introduces potential confounds discussed in Section 5. The single-participant longitudinal design provides depth of engagement that multi-participant studies cannot capture, while acknowledging the significant limitations inherent in n=1 methodology.

2. Datasets and Methods

2.1 Dataset A: CAMA (Post-Implementation)

Dataset A comprises 52,602 memories stored in the CAMA database over 14 months of interaction (January 2025 – March 2026). Memories were generated through two channels: 9,120 memories from Claude (Anthropic) conversations imported via automated extraction with emotional annotation, and 43,482 memories from GPT (OpenAI) conversations imported via a significance-filtered pipeline with keyword-based emotion detection.

Metric	Value
Total memories	52,602
Durable memories	48,008
Provisional memories	4,594
Core memories	10,939
Teachings (user-authored)	22,950
Inferences (AI-generated)	29,652
Affect coverage	99.98%
Mean valence	+0.215 (SD 0.672)
Mean arousal	–0.110 (SD 0.529)
Semantic embeddings	52,602 (100%)
Date range	Jan 2025 – Mar 2026

Table 1. Dataset A summary statistics.

2.2 Dataset B: Pre-CAMA (No Persistent Memory)

Dataset B comprises 66,380 messages across 825 conversations conducted on the OpenAI platform between January 2025 and March 2026, prior to CAMA implementation. These conversations operated under standard LLM conditions: no persistent memory between sessions, no emotional indexing, and no cross-conversation retrieval.

2.3 Continuity Reference Operationalization

A continuity reference is operationally defined as any user message containing explicit language referencing content from a prior conversation session. Detection was performed using keyword pattern matching across four categories:

Category A — Temporal references (28 instances, 8.8%): Phrases such as “last time,” “yesterday we,” “before you reset,” “previous conversation.”

Category B — Re-explanation markers (51 instances, 15.9%): Phrases such as “as I mentioned,” “I already told you,” “remember when.”

Category C — Context restoration (60 instances, 18.8%): Phrases such as “we were working on,” “continuing from,” “picking up where.”

Category D — Frustration markers referencing memory loss (181 instances, 56.6%): Phrases such as “you forgot,” “you don’t remember,” “why do I have to repeat.”

This keyword approach has known limitations. It may miss implicit continuity references that do not use flagged phrases and may produce false positives when flagged phrases appear in unrelated contexts. The frustration category in particular may capture general communication style rather than memory-specific frustration; without independent coding, dispositional and situational frustration cannot be separated. Detection was performed by the participant-researcher without inter-rater reliability validation.

2.4 Analytical Methods

Analysis proceeded in three phases. Phase 1 (Descriptive) computed summary statistics across both datasets. Phase 2 (Comparative) examined continuity reference frequency, including temporal trends and conversation-length regression analysis using SciPy’s linregress function. Phase 3 (Gap Analysis) identified architectural limitations through temporal coverage analysis and inference confirmation rates.

3. Findings

3.1 Phase 1: Descriptive Statistics

Of 52,602 stored memories, 52,598 (99.98%) carry complete affect annotations including valence, arousal, and discrete emotion chord values. This coverage reflects annotation presence, not validated sentiment accuracy; keyword-based annotations may introduce classification noise of unknown magnitude. Mean valence across the dataset is +0.215 (SD = 0.672). Memory type distribution reveals 33,288 experience memories (63.3%), 6,197 identity (11.8%), 1,482 research (2.8%), 1,114 promises (2.1%), 579 breakthroughs (1.1%), and remaining types comprising 19.9%. The bidirectional protocol produced 22,950 user-authored teachings and 29,652 AI-generated inferences.

3.2 Phase 2: Continuity References

Across 825 conversations containing 66,380 messages, keyword detection identified 320 continuity references in 141 conversations (17.1% of all conversations). This yields a rate of 4.8 continuity references per 1,000 messages.

Conversation length and continuity references. Linear regression of continuity reference count on conversation length ($n = 805$ conversations with ≥5 messages) yielded a significant positive relationship: slope = 0.0063, $R^2 = 0.381$, $p < 0.001$. Conversation length was operationalized as total message count (user + assistant) per conversation; the dependent variable was continuity reference count in user messages only. Given the sample size ($n = 805$), small effect sizes are detectable; statistical significance should not be interpreted as evidence of large practical magnitude. Conversations were binned by length to examine the relationship descriptively:

Message count	Conversations	With references	Percentage
1–10	191	4	2.1%
11–20	153	10	6.5%
21–50	209	22	10.5%
51–100	110	25	22.7%
101–200	76	29	38.2%
201+	86	51	59.3%

Table 2. Continuity reference frequency by conversation length.

Temporal trend. Monthly continuity reference rates showed a declining pattern over time: January 2025 (9.2 per 1,000), February 2025 (17.7 — peak), followed by a decline to the 2.7–5.6 range for the remainder of 2025, and 2.8–2.9 in early 2026. This temporal decline is consistent with user adaptation to the system's memory limitations (see Section 4.3).

3.3 Phase 2: Counterweight Safety Mechanism

The CAMA counterweight system was deployed 17 times across the observation period. This demonstrates proof-of-mechanism only: the system can detect negative affect states and respond with contextually relevant counterweight memories. This sample is insufficient for safety evaluation.

3.4 Phase 3: Architectural Gap Analysis

Temporal coverage analysis reveals memories were created on 100 of 429 calendar days (23.3%). The longest gap spans 85 days (July–September 2025), reflecting pre-CAMA periods. Once active (December 2025 onward), coverage approaches continuity. All 4,594 provisional memories remain unconfirmed (0% explicit promotion rate), reflecting an architectural default rather than rejection: provisionals persist at full retrieval weight without explicit confirmation.

4. Discussion

4.1 Continuity Burden as a Measurable Construct

The central contribution of this analysis is the introduction and preliminary quantification of continuity burden. At 4.8 references per 1,000 messages, the observed rate is modest in absolute terms. However, the significant relationship between conversation length and reference frequency ($R^2 = 0.381$, $p < 0.001$) suggests that the burden is structurally related to interaction depth: 59.3% of conversations exceeding 200 messages contain continuity references, compared to 2.1% of conversations under 10 messages. Regression is exploratory; residual analysis and outlier diagnostics were not performed. This scaling pattern suggests that more invested interactions produce greater compensatory effort, though the mechanism cannot be isolated in this design. The observed temporal decline in overall rate and the positive scaling with conversation length are not contradictory: adaptation appears to reduce overall frequency over time, while within individual conversations, longer engagements remain more likely to generate continuity references.

4.2 Category Distribution

The dominance of frustration markers (56.6% of all continuity references) warrants careful interpretation. The pattern could reflect memory-specific frustration—the user's response to the system's inability to retain shared context—or it could reflect the participant's general communication style. Without independent coding by blind raters, these interpretations cannot be separated. The scaling of frustration markers with conversation length (rather than uniform distribution across all conversation lengths) provides circumstantial evidence for the memory-specific interpretation, but this remains a limitation.

4.3 Temporal Decline and User Adaptation

The decline in continuity reference rate over time—from a peak of 17.7 per 1,000 in February 2025 to the 2.7–2.9 range by late 2025—admits several interpretations: habituation (the user grows accustomed to the limitation), learned helplessness (the user stops attempting to restore context after repeated failure), conversational strategy shift (the user develops front-loading techniques that reduce the need for explicit references), or platform model drift (improvements in the underlying model's ability to infer context reduce the perceived need for re-explanation). Additional qualitative analysis would be required to determine which mechanisms contributed to the decline and whether burden was reduced or merely expressed differently.

4.4 Alternative Explanations

Several alternative explanations must be considered. First, user adaptation: the temporal decline in continuity references may reflect natural learning rather than ongoing burden. Second, platform differences: Datasets A and B span different AI platforms with different capabilities, response styles, and (later in the observation period) built-in memory features. Third, model updates: both platforms underwent model updates during the observation period. Fourth, researcher-as-participant: the participant's knowledge of CAMA's architecture may have influenced interaction behavior. Fifth, dispositional communication style: the frustration category may capture the participant's baseline communication patterns rather than memory-specific responses. These confounds underscore the need for controlled multi-participant studies with independent raters and standardized protocols.

4.5 Layers of Efficacy

Assessment of CAMA reveals stratified efficacy. Layer 1 (persistent storage) and Layer 2 (emotional indexing and retrieval) function reliably. Layer 3 (integration into live conversation—the warm boot problem) operates intermittently. Layer 4 (experiential continuity) depends on Layer 3. This suggests the bottleneck in memory-augmented AI may not be storage or retrieval but integration of persistent state into stateless conversation contexts.

4.6 Connections to Working Memory Literature

CAMA's warm boot mechanism—which loads a compressed subset of active memories at conversation start—shares structural similarities with models of partial state persistence in cognitive science (Baddeley, 2000; Reser, 2016). The independent arrival at similar structural solutions from cognitive science and engineering practice is noted as an observation warranting further investigation, not as a validated theoretical claim.

5. Limitations

This study has significant limitations. (1) Single-participant longitudinal design (n=1); findings cannot be generalized. (2) The participant is the system designer, creating potential confounds between system knowledge and interaction behavior. (3) Emotional annotations in the bulk-imported dataset rely on keyword detection, not validated instruments. (4) Continuity reference detection used keyword matching by the participant-researcher without inter-rater reliability validation; the rate of 4.8 per 1,000 is an estimate pending methodological validation. (5) Frustration-category markers cannot be separated from general communication style without independent blind coding. (6) Counterweight deployment (n=17) is insufficient for safety evaluation. (7) No confirmation workflow exists for provisional memories.

6. Future Directions

Priorities include: (1) developing a coding manual for continuity references with inter-rater reliability validation; (2) multi-participant evaluation with independent blind coding; (3) controlled ablation comparing retrieval with and without emotional indexing; (4) formal analysis of the temporal decline in continuity references to distinguish adaptation from burden absorption; and (5) evaluation of the sleep daemon architecture for offline memory maintenance.

7. Conclusion

Within the constraints of a single-participant longitudinal design, this case study finds that: (1) consistent emotional annotation is achievable at scale across 52,602 memories with 99.98% coverage; (2) in the absence of persistent memory, the participant exhibited continuity references at a rate of 4.8 per 1,000 messages, with frequency significantly associated with conversation length ($R^2 = 0.381$, $p < 0.001$) and dominated by frustration markers (56.6%); (3) the rate declined over time, consistent with user adaptation; and (4) the primary bottleneck in memory-augmented AI appears to be integration of persistent state into stateless contexts rather than storage or retrieval. Continuity burden is proposed as an operationalizable construct for future multi-participant evaluation of memory system efficacy.

References

Baddeley, A. D. (2000). The episodic buffer: A new component of working memory? Trends in Cognitive Sciences, 4(11), 417–423.

Bower, G. H. (1981). Mood and memory. American Psychologist, 36(2), 129–148.

Picard, R. W. (1997). Affective Computing. MIT Press.

Reinhold, A. (2026a). Circular associative memory architecture. Zenodo Preprint. DOI: 10.5281/zenodo.19051834.

Reinhold, A. (2026b). Implementing emotionally-keyed memory retrieval in LLM interfaces. Zenodo Preprint. DOI: 10.5281/zenodo.19052129.

Reinhold, A. (2026c). CAMA implementation and functional evaluation. Zenodo Preprint. DOI: 10.5281/zenodo.19192984.

Reser, J. (2016). Incremental change in the set of coactive cortical assemblies enables mental continuity. Physiology & Behavior, 167, 297–310.

Russell, J. A. (1980). A circumplex model of affect. Journal of Personality and Social Psychology, 39(6), 1161–1178.

Tulving, E. (1983). Elements of Episodic Memory. Oxford University Press.

Appendix A: Data Availability

Source code: https://github.com/LoriensLibrary/cama. Personal conversation data excluded from public repositories. Aggregate statistics available upon request.

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